Adricore SMS Gateway — Sasula Collections API

Audience: Sasula / Service Cops integration team
Version: 1.0
Date: 8 September 2026

This document describes the APIs Sasula must call to look up a gateway partner (customer reference) and to post a successful collection. Payments credit the partner's SMS wallet.

---

1. Overview

Method	Purpose
Login	Obtain a JWT for the dedicated Sasula integration user
Query customer	Validate the customer reference (partner code) and return name, outstanding amount, and partner details
Post payment	Record a Sasula collection: top up the partner credit account and write a credit transaction

All requests are `POST` with `Content-Type: application/json`. All Sasula business methods require a Bearer token.

Customer reference is the unique partner code issued by Adricore, for example `ADRS100002`.

---

2. Base URL

Replace the host with the environment you are given.

```
{BASE_URL}/api/v1
```

Examples:

Environment	Base URL
UAT	`http://{{uat-host}}/api/v1`
Production	`https://{production-host}/api/v1`

---

3. Authentication

Adricore will create a dedicated integration user (role **Sasula Integration**) and share the username and password securely. That user can only call these collection APIs.

3.1 Login

URL: `POST {BASE_URL}/Auth/login/`

No `Authorization` header.

Request

Field	Type	Required	Description
`username`	string	Yes	Integration username
`password`	string	Yes	Integration password

```
{
  "username": "sasula",
  "password": "{provided-password}"
}
```

Success (`returnCode = 0`)

```
{
  "returnCode": 0,
  "returnMessage": "Welcome back sasula",
  "returnObject": {
    "token": "eyJhbGciOiJIUzI1NiJ9...",
    "user": {
      "username": "sasula",
      "roleCode": "SASULA"
    }
  }
```

```
}
```

Use `returnObject.token` on every subsequent request:

```
Authorization: Bearer {token}
```

Notes

- Tokens are valid for **12 hours**.
- Logging in again issues a new token and **invalidates** the previous one. Keep a single token in use until it expires, then login again.
- Expired or missing tokens return `returnCode` `401` with `returnMessage` `AUTHENTICATION REQUIRED` or `TOKEN EXPIRED`.

3.2 Calling authenticated APIs

```
POST {BASE_URL}/Sasula/{action}/
Authorization: Bearer {token}
Content-Type: application/json
```

---

4. Standard response envelope

Every response is JSON in this shape:

Field	Type	Description
`returnCode`	number	`0` = success. Any other value is a failure.
`returnMessage`	string	Human-readable result. On failure, this is the reason.
`returnObject`	object or `null`	Payload on success; usually `null` on failure.

HTTP status is typically `200`. Always inspect `returnCode`, not only HTTP status.

Return codes

`returnCode`	Meaning
`0`	Success
`1`	Business validation error (unknown customer, inactive customer, bad amount, duplicate receipt, etc.)
`401`	Missing, invalid, or expired token
`403`	Authenticated but the user is not allowed to call Sasula APIs
`500`	Unexpected server error (`returnMessage` has detail)

---

5. Query customer reference details

Validates the partner code and returns the details Sasula needs to present the collection.

URL: `POST {BASE_URL}/Sasula/queryCustomer/`

Headers: `Authorization: Bearer {token}`

Input

Field	Type	Required	Description
`customerReference`	string	Yes	Unique customer reference (Adricore partner code), e.g. `ADRS100002`

`partnerCode` is accepted as an alias of `customerReference`.

```
{
  "customerReference": "ADRS100002"
}
```

Output (`returnObject`)

Field	Type	Description
`customerReference`	string	Echo of the partner code
`customerName`	string	Partner / organisation name
`amountOutstanding`	number	Sum of **pending** billing invoices. `0` if none
`fixedAmount`	boolean	`true` if outstanding is greater than zero (pay that amount in full). `false` if outstanding is `0` (any top-up amount is allowed)
`businessReference`	string	Same as the partner code; identifies the wallet the collection credits

Field	Type	Description
`partnerId`	string (UUID)	Internal partner id
`partnerCode`	string	Same as `customerReference`
`email`	string	Partner email, if set
`phone`	string	Partner phone, if set
`status`	string	`active`
`currency`	string	Wallet currency, e.g. UGX

Success example

```
{
  "returnCode": 0,
  "returnMessage": "Customer found",
  "returnObject": {
    "customerReference": "ADRS100002",
    "customerName": "Example Partner Ltd",
    "amountOutstanding": 0,
    "fixedAmount": false,
    "businessReference": "ADRS100002",
    "partnerId": "e87ca817-050a-4965-a9fe-254cfeb35314",
    "partnerCode": "ADRS100002",
    "email": "billing@example.com",
    "phone": "+256700000000",
    "status": "active",
    "currency": "UGX"
  }
}
```

Outstanding amount

- Taken from pending rows in `billing_invoices` for that partner.
- If the partner has no pending invoices, `amountOutstanding` is `0` and `fixedAmount` is `false`.
- If there are pending invoices, `amountOutstanding` is their total and `fixedAmount` is `true`.

Failure examples

Situation	`returnCode`	`returnMessage`
Missing reference	1	Customer reference cannot be empty
Unknown code	1	Unknown customer reference
Inactive partner	1	Customer reference is inactive
No / bad token	401	AUTHENTICATION REQUIRED

---

6. Post payment

Called after Sasula has collected funds. The gateway:

1. Validates the customer reference (active partner).
2. Rejects duplicate Sasula receipt numbers.
3. Credits the partner's credit account by `amountPaid`.
4. Inserts a `TOPUP` credit transaction (receipt stored as the transaction reference).

URL: `POST {BASE_URL}/Sasula/postPayment/`

Headers: `Authorization: Bearer {token}`

Input

Field	Type	Required	Description
`customerReference`	string	Yes	Partner code, e.g. `ADRS100002`
`amountPaid`	number or string	Yes	Amount collected. Must be greater than zero
`sasulaReceiptNumber`	string	Yes	Unique receipt on Sasula. Used for duplicate detection
`paymentChannel`	string	No	Where payment was made, e.g. `Centenary Bank`, `MTN mobile money`
`thirdPartyTransactionReference`	string	No	Transaction id in Sasula / third party
`channelTransactionId`	string	No	Transaction id on the source channel
`datePaid`	string	No	Date/time paid (ISO-8601 recommended)
`transactionDescription`	string	No	Narrative
`payerNumber`	string	No	Payer MSISDN or account, if the channel provides it
`payerName`	string	No	Payer name, if the channel provides it

```
{
  "customerReference": "ADRS100002",
  "amountPaid": 50000,
  "paymentChannel": "MTN mobile money",
  "thirdPartyTransactionReference": "TP-123456",
  "channelTransactionId": "CH-987654",
  "sasulaReceiptNumber": "SAS-20260908-0001",
  "datePaid": "2026-09-08T14:30:00",
  "transactionDescription": "SMS wallet top-up",
  "payerNumber": "256700000000",
  "payerName": "Jane Okello"
}
```

Output (`returnObject`)

Field	Type	Description
`customerReference`	string	Partner code
`customerName`	string	Partner name
`amountPaid`	number	Amount credited
`balanceAfter`	number	Wallet balance after this top-up
`transactionId`	string (UUID)	Gateway credit-transaction id
`sasulaReceiptNumber`	string	Echo of the receipt
`reference`	string	Stored transaction reference (the Sasula receipt)
`currency`	string	Wallet currency

Success example

```
{
  "returnCode": 0,
  "returnMessage": "Payment posted",
  "returnObject": {
    "customerReference": "ADRS100002",
    "customerName": "Example Partner Ltd",
    "amountPaid": 50000,
    "balanceAfter": 75000,
    "transactionId": "0471674b-00ab-4c94-93d8-8a96d8a3dd6c",
    "sasulaReceiptNumber": "SAS-20260908-0001",
    "reference": "SAS-20260908-0001",
    "currency": "UGX"
  }
}
```

Duplicates

`sasulaReceiptNumber` must be unique. Posting the same receipt again returns:

```
{
  "returnCode": 1,
  "returnMessage": "Duplicate Sasula receipt number",
  "returnObject": null
}
```

Treat that as already processed: do not retry with a new receipt for the same collection.

Other failures

Situation	`returnCode`	`returnMessage`
Missing reference	1	Customer reference cannot be empty
Unknown / inactive partner	1	Unknown customer reference / Customer reference is inactive
Missing amount	1	Amount paid cannot be empty
Amount ≤ 0	1	Amount paid must be greater than zero
Non-numeric amount	1	Amount paid must be a number
Missing receipt	1	Sasula receipt number cannot be empty
No / bad token	`401`	AUTHENTICATION REQUIRED

---

7. Suggested flow

1. Login once; cache the token until it expires (12 hours).
2. On enquiry: **Query customer** with the partner code the payer quoted.

3. If `returnCode` is not `0`, show `returnMessage` and stop.
4. Present `customerName`, `amountOutstanding`, and `currency`.
    - If `fixedAmount` is `true`, collect exactly `amountOutstanding`.
    - If `fixedAmount` is `false`, collect any amount the payer chooses (wallet top-up).
5. After a successful collection: **Post payment** with the Sasula receipt number and channel details.
6. If post payment returns duplicate receipt, treat as success (already posted).
7. On `401`, login again and retry the same business request **once**.

---

8. cURL examples

Replace `{BASE_URL}`, `{username}`, `{password}`, and `{token}`.

Login

```
curl -s -X POST "{BASE_URL}/Auth/login/" \
  -H "Content-Type: application/json" \
  -d '{"username":"{username}","password":"{password}"}'
```

Query customer

```
curl -s -X POST "{BASE_URL}/Sasula/queryCustomer/" \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer {token}" \
  -d '{"customerReference":"ADRS100002"}'
```

Post payment

```
curl -s -X POST "{BASE_URL}/Sasula/postPayment/" \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer {token}" \
  -d '{
    "customerReference": "ADRS100002",
    "amountPaid": 50000,
    "paymentChannel": "MTN mobile money",
    "thirdPartyTransactionReference": "TP-123456",
    "channelTransactionId": "CH-987654",
    "sasulaReceiptNumber": "SAS-20260908-0001",
    "datePaid": "2026-09-08T14:30:00",
    "transactionDescription": "SMS wallet top-up",
    "payerNumber": "256700000000",
    "payerName": "Jane Okello"
  }'
```

---

9. Alternate request style

The gateway also accepts a single endpoint with `SERVICE` and `ACTION` in the body:

```
POST {BASE_URL}/
```

```
{
  "SERVICE": "Auth",
  "ACTION": "login",
  "username": "sasula",
  "password": "{provided-password}"
}
```

```
{
  "SERVICE": "Sasula",
  "ACTION": "queryCustomer",
  "customerReference": "ADRS100002"
}
```

```
{
  "SERVICE": "Sasula",
  "ACTION": "postPayment",
  "customerReference": "ADRS100002",
```

```
  "amountPaid": 50000,
  "sasulaReceiptNumber": "SAS-20260908-0001"
}
```

The path style in sections 3–6 is preferred.

---

10. Credentials

Adricore operations will provision:

Item	Value
Base URL	*to be provided per environment*
Username	*dedicated Sasula integration user*
Password	*shared out of band*
Role	SASULA (Sasula Integration)

Do not use operator or partner-admin accounts. Rotate the password if it is exposed. Contact Adricore if tokens fail with `403 NOT AUTHORISED`.